HyProC - Manual

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1 Getting started

HyProC makes use of SRIM [1] to calculate stopping powers. It thus has to be pre-installed for the code to work, and its path must be specified in the settings.json file via the "SRIM_path" parameter.

From the HyProC folder, run the UI.py file to launch the code.

2 Button behaviours

Import excitation curve	Loads an excitation curve. To get data, the code will look for the "Energie (keV)", "Ng/µC" and "Incertitude" headers and read the following rows on the relevant columns. The headers names can be changed by editing the settings.json file (no need to close/restart the program). File format: Excel (.xlsx, .xls, .csv) Having the file opened in Excel when loading will cause an error.

Run calculation	Simulates an excitation curve, i.e. calculates the yield for each energy value of the loaded excitation curve. Thus, an excitation curve has to be loaded for the function to run.
Extract H profile	Plots the hydrogen depth profile based on the current target.
Load target	Self-explanatory.
Save target	File format: .json
Save simulated curve data	Saves the data of the simulated excitation curve in a two-column tab-separated .txt file.
(Standard) Load	Self-explanatory. Since a standard should be uniform, it should contain only one layer. Thus when loading a multi-layer target as standard, the program will ask which layer to keep. The others will NOT be loaded. File format: .json.
(Standard) Save	
Scroll Up/Down	Self-explanatory.
Add/Remove Layer Move Up/Down Duplicate Layer	Self-explanatory.
Add/Remove element	Self-explanatory.
Isolate & Normalize	Rescales atomic percentages in a layer so they add up to 100%, while keeping the selected element's percentage fixed. It proportionally adjusts the other elements' values to maintain balance. This is especially useful when incorporating an element into a compound (e.g. hydrogen in SiO_2). Define the compound with the correct atomic concentrations, add the new element with the desired concentration, then hit the button.

3 Physics

3.1 Stopping powers

HyProC calculates stopping powers ($S = \frac{-dE}{dx}$) through SRIM [1] via the following. It considers an incident ^{15}N particle who energy is equal to the loaded excitation curve's maximum energy. This energy E_0^{In} is used to calculate the energy the particle will have

when exiting the first layer:

$$E_0^{\mathrm{Out}}[\mathrm{keV}] = E_0^{\mathrm{In}} - S(E_0^{\mathrm{In}})\, x \tag{1}$$

where x represents the areal density of the layer (in TFU[1]). $S(E_0^{\mathrm{Out}})$ is then computed. Finally, the stopping power attributed to the layer is given by:

$$S[\mathrm{keV/TFU}] = \frac{1}{2}\left(S(E_0^{\mathrm{Out}}) + S(E_0^{\mathrm{In}})\right) \tag{2}$$

This means the stopping power is constant by layer. This is of course an approximation. To keep it relevant, the code checks whether the difference between E_0^{Out} and E_0^{In} is inferior to 0.5%. If the test is passed, formula (2) is used. If not, the layer is split in 2, stopping powers are recalculated and the test is taken again. If the test fails again, the layer is no longer split in 2 but in 3. This process is repeated unless the condition is met. Each iteration increases the number of daughter layers by one.

3.2 Beam width

An ion beam is rarely mono-energetic, even before it reaches the target. To reflect this, HyProC assumes a gaussian energy distribution of the beam. The user can specify the standard deviation of this gaussian distribution in the text field next to "Beam width SD (keV)". If set to 0, the beam is considered mono-energetic.

3.3 Doppler broadening

A Doppler effect causes a gaussian broadening of the reaction's resonance [2]. This can be taken into account or not using the tickbox in the Options section. The standard deviation of this gaussian broadening can be calculated using:

$$\delta_D[\mathrm{keV}] = \sqrt{\frac{2 m_N E_H E_R}{m_H}} \tag{3}$$

where m_N, m_H, E_H and E_R are the nitrogen and hydrogen mass, the H zero-point vibrational energy and the resonance energy respectively. While m_N, m_H and E_R are very well known, this cannot be said for E_H. Data are available in the literature for Si [3], Ti [4] and Pt [3]. If the matrix containing hydrogen is made of one of those 3 elements, the literature data are used. Otherwise, the following linear regression is used

$$\delta_D[\mathrm{keV}] = -0.0218\, Z + 4.24 \tag{4}$$

where Z is the effective atomic number of the material (ie a Bragg rule is used in case of compounds).

3.4 Straggling

Statistical variations in the number of collisions between the projectiles and the target's electrons lead to a broadening of the energy distribution of the beam as it penetrates

[1] 1 TFU (Thin Film Units) = 10^{15} at./cm^2

the target. To describe this, HyProC considers a gaussian broadening whose standard deviation can be obtained using Rud's empirical formula [5]:

$$\delta_S(x)[\text{keV}] = \frac{2.03}{2.355} Z^{0.39} \sqrt{x} \tag{5}$$

where Z is the atomic number of the material and x its areal density in 10^{16} at./cm^2.

However during testing, straggling was found to be overestimated. Results obtained using the Rud formula were then compared with SRIM outputs, revealing a discrepancy factor of 1.7. A second straggling model "Rud corr" was thus implemented in HyProC:

$$\delta_S(x)[\text{keV}] = \frac{1}{1.7} \frac{2.03}{2.355} Z^{0.39} \sqrt{x} \tag{6}$$

Bohr's straggling model is also implemented [6]:

$$\delta_S(x)[\text{keV}] = \sqrt{0.260532\ Z_{\text{beam}}^2 Z x} \tag{7}$$

with x in 10^{18} at./cm^2

For compounds, the straggling variance of each individual element must be summed to get the total variance. For instance [7]:

$$\delta_{\text{SiO}_2}^2(x)[\text{keV}^2] = \frac{1}{3}\,\delta_{\text{Si}}(x)^2 + \frac{2}{3}\,\delta_{\text{O}}(x)^2 \tag{8}$$

Note that for small energy loss and thin layers, the straggling isn't gaussian. The distribution is asymmetric and follows Vavilov's theory [8]. This is currently not implemented in HyProC.

4 Calculations

4.1 Results: Chi-square test

When a simulated excitation curve is generated, a chi-square test is made to assess how well it fits the loaded excitation curve. The value displayed on the "Chi-square history" graph is obtained with this formula:

$$\chi^2 = \frac{1}{N} \sum_{i=0}^{N} (y_{\text{exp},i} - y_{\text{sim},i})^2$$

where $y_{\text{exp},i}$ and $y_{\text{sim},i}$ represent the experimental and simulated yields of the i-th datapoint whilst N is the number of datapoints.

4.2 Broadening data

Broadening data can be saved by ticking the "Save broadening data" box. When calculations are made with this option on, folders will be created using this architecture:

```
C:/HyProC/
└── Session/
    └── Run/
        └── Datapoint/
            ├── Beam.txt
            ├── Stragg.txt
            ├── Doppler.txt
            ├── Total_Gauss.txt
            ├── xsec.txt
            ├── Total_Broadening.txt
            └── Total_Broadening_TFU.txt
```

Total_Gauss represents the convolution of all gaussian broadening (Beam, Straggling and Doppler). Convolving this curve with the cross section of the reaction (xsec, modelled as a lorentzian curve) gives a Voigt profile representing the total broadening (Total_Broadening file). Total_Broadening_TFU represents the same data with a different x-axis (areal density in TFU instead energy in keV).

A Session folder is created when broadening data are saved or that the track target changes option is enabled for the first time after the UI is launched. A Run folder is created whenever a calculation is made (with the save broadening data on or with the target tracking on). For each incident energy, a datapoint folder is created.

The broadening data can be visualised using the broadplotter.py script (recommended) or through a third-party software (OriginLab, ...).

4.3 Track target change

By ticking the Track target change box, the code will save the target each time a simulated excitation curve is generated. The target will be saved in the "Run" folder (c.f. tree above).

References

[1] James F Ziegler, Matthias D Ziegler, and Jochen P Biersack. "SRIM–The stopping and range of ions in matter (2010)". In: *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms* 268.11-12 (2010), pp. 1818–1823.

[2] KM Horn and WA Lanford. "Observation of the bond-dependent Doppler broadening of the p (15N, $\alpha\gamma$) 12C nuclear reaction". In: *Nuclear Instruments and Methods in Physics Research Section B: Beam Interactions with Materials and Atoms* 29.4 (1988), pp. 609–620.

[3] K Fukutani et al. "Zero-point vibration of hydrogen adsorbed on Si and Pt surfaces". In: *Physical review letters* 88.11 (2002), p. 116101.

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[7] Yongqiang Wang and Michael Nastasi. *Handbook of modern ion beam materials analysis*. MRS, Materials Research Soc., 2010.

[8] PV Vavilov. "Ionization losses of high-energy heavy particles". In: *Soviet Phys. JETP* 5 (1957).